

Exploring the Impact of Socioeconomic Factors on Crime Rates in the U.S. (2010–2019)

Data Visualization for Story Telling

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1. Introduction

Overview of the Research Area

Crime remains a persistent challenge in the United States, influencing communities, public safety, and economic stability. One area of ongoing research is the relationship between crime rates and various socioeconomic factors, including unemployment and income inequality.

Understanding these connections is crucial for shaping effective public policy and developing targeted interventions that promote safety and social well-being.

This project explores the relationship between socioeconomic indicators and crime rates across U.S. states from 2010 to 2019. By integrating FBI crime data with employment and income statistics from the U.S. Census, the analysis investigates how violent and property crime trends relate to patterns in unemployment, household income, gender, age, and regional dynamics. The project also examines differences in crime prevalence across income groups and evaluates how specific crime types have evolved over time.

The goal of the study is to uncover meaningful patterns and correlations that provide insight into the structural and economic factors associated with crime. By analyzing trends at both national and state levels, the research contributes to a broader understanding of how economic and demographic contexts shape crime outcomes in the United States.

2. Methodology

➤ FBI Uniform Crime Reporting (UCR) Program Data (2010–2019)

- **Source:** FBI UCR - <https://www.fbi.gov/how-we-can-help-you/more-fbi-services-and-information/ucr/publications>

- **Description:** This dataset includes detailed information on various types of crime, including violent crimes (such as murder, rape, assault, and robbery) and property crimes (such as burglary, larceny-theft, and motor vehicle theft) reported by law enforcement agencies across the U.S. between 2010 and 2019. It serves as the primary data for understanding crime trends and analyzing correlations with socioeconomic factors like unemployment and income inequality. The UCR data is broken down by state, allowing for comparison across regions.

➤ **U.S. Census Bureau Data**

- **Source:** U.S. Census Bureau - <https://data.census.gov/>
- **Description:**
 - **S2301 Employment Status (2010–2019):** This dataset provides detailed information on the unemployment rate and employment status across different regions of the U.S. It is used to examine how variations in unemployment levels across states may be associated with differences in crime rates.
 - **S1901 Income in the Past 12 Months (2010–2019):** This dataset provides information on income levels by region and is used to measure income inequality across states. It helps identify areas with varying levels of income disparity and examines how those differences may be associated with differences in crime rates across states.

The combination of these three datasets enabled an integrated analysis of how socioeconomic factors, such as unemployment and income inequality, may be associated with crime rate patterns

across different U.S. states. The datasets were joined using state and year to support effective cross-comparison and visualization.

3. Analysis

1. Data Preparation

To prepare the datasets for meaningful analysis and visualization, preprocessing was conducted in both Python and Tableau.

Python was used to perform the following preprocessing tasks to standardize and structure the data for integration and visualization:

✓ Data Extraction and Cleanup:

1. The original datasets were segmented by year; they were merged into single, year-wise consolidated files for each theme (crime, employment, income)
2. For *crime data*, .xls files were converted to .xlsx, then parsed to extract only the “State Total” rows for each year and state. Crime counts were normalized per 100,000 population to allow meaningful comparisons.
3. *Unemployment data* from the Census S2301 table was cleaned to include only U.S. states.
4. Column names were standardized, commas in numeric values were removed, and values were coerced into proper numeric types.
5. *Income data* from the Census S1901 table was filtered, cleaned, and structured for analysis.
6. For *unemployment* and *income* data, only files ending with `\*-Data.csv` were extracted.
7. *All datasets* were filtered to include only the 50 U.S. states, with state names cleaned (e.g., capitalization, removal of numbers and punctuations).

8. Numeric values were properly converted and missing values handled.

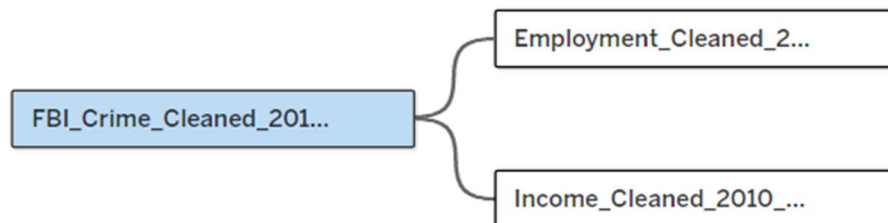
✓ **Data Consistency and Integrity:**

1. All three datasets were standardized in format and column structure.
2. Non-state regions such as "Puerto Rico" and "District of Columbia" were excluded to
3. maintain consistency across datasets.
4. Spurious rows (example: "Geographic Area Name") and malformed state names (example:ALABAMA1, NEW YORK6, etc.) were successfully cleaned.
5. Each dataset includes state and year columns, ready to be used for Tableau joins.

After preprocessing, three cleaned CSV datasets: FBI Crime, Employment, and Income, were loaded into Tableau for further analysis and visualization.

In **Tableau**, further enhancements supported interpretability and analysis:

1. Joined the three datasets using State and Year fields as keys for relational linking.



2. Created calculated fields such as gender unemployment gap and three-year moving averages.
3. Built parameters and filters for dynamic selection (e.g., age groups, crime types, regions).

4. Incorporated region and state-level breakdowns to support geographic comparisons in the visual analysis.
5. A pivot transformation was applied to restructure unemployment data by race. The race categories were extracted into a single *Race* dimension with corresponding *Unemployment by Race* values, allowing comparison across racial groups.

2. Tools Used

- **Python:** Used for initial data extraction, cleaning and preprocessing prior to visualization.
- **Tableau:** Primary tool for data visualization and dashboard creation.

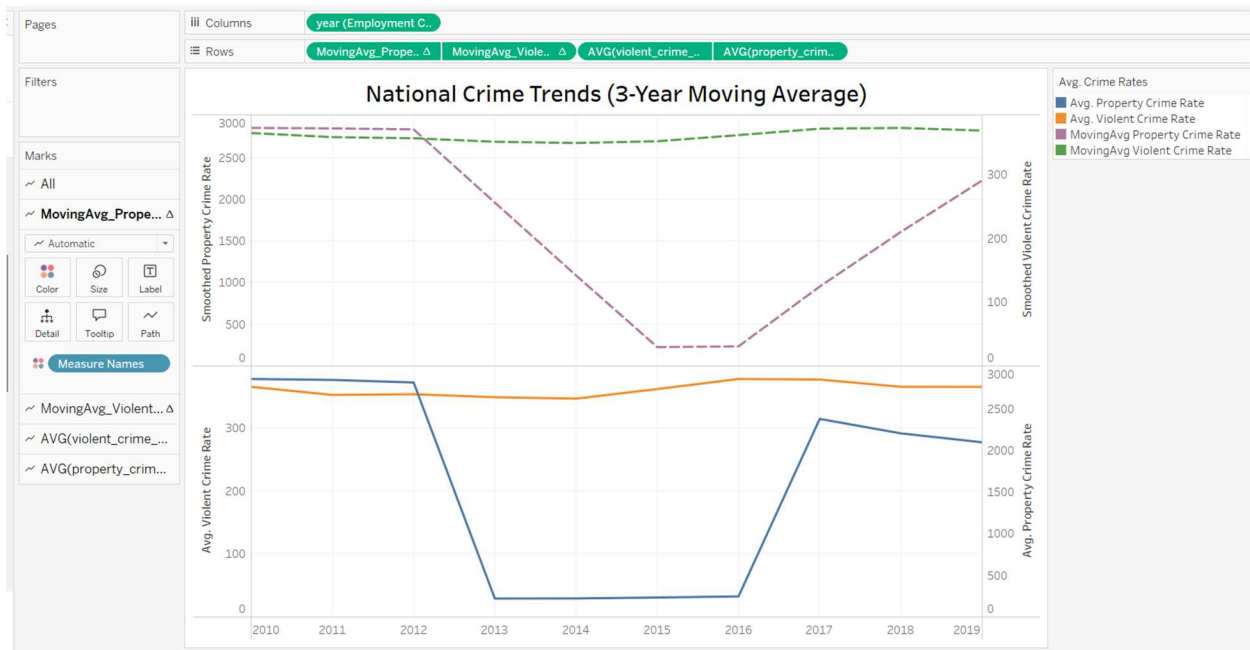
3. Data Visualization

The visualizations developed in Tableau offer structured insights into national and regional crime patterns, as well as the socioeconomic variables influencing them. By drawing on multiple data sources and segmenting by geography, age, income, and gender, the analysis answers key research questions concerning trends, disparities, and associations between crime and socioeconomic factors. The following sections detail how the visual evidence supports each research question.

✓ Research Questions:

1. How have national violent and property crime rates changed from 2010 to 2019?

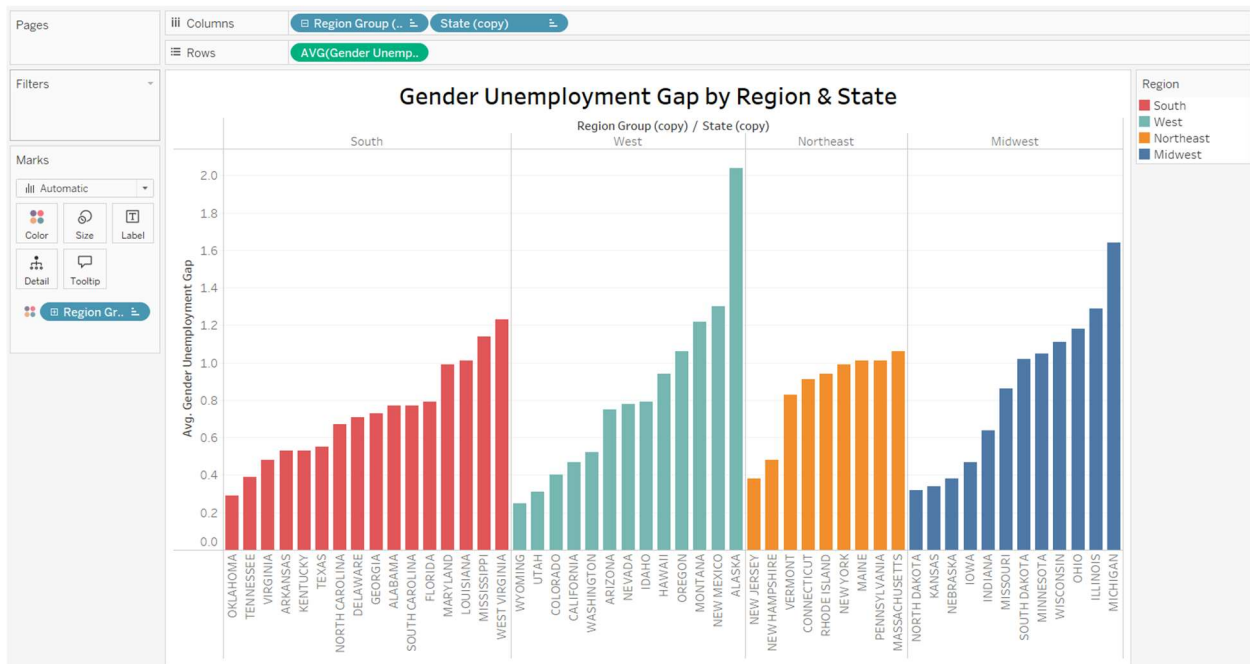
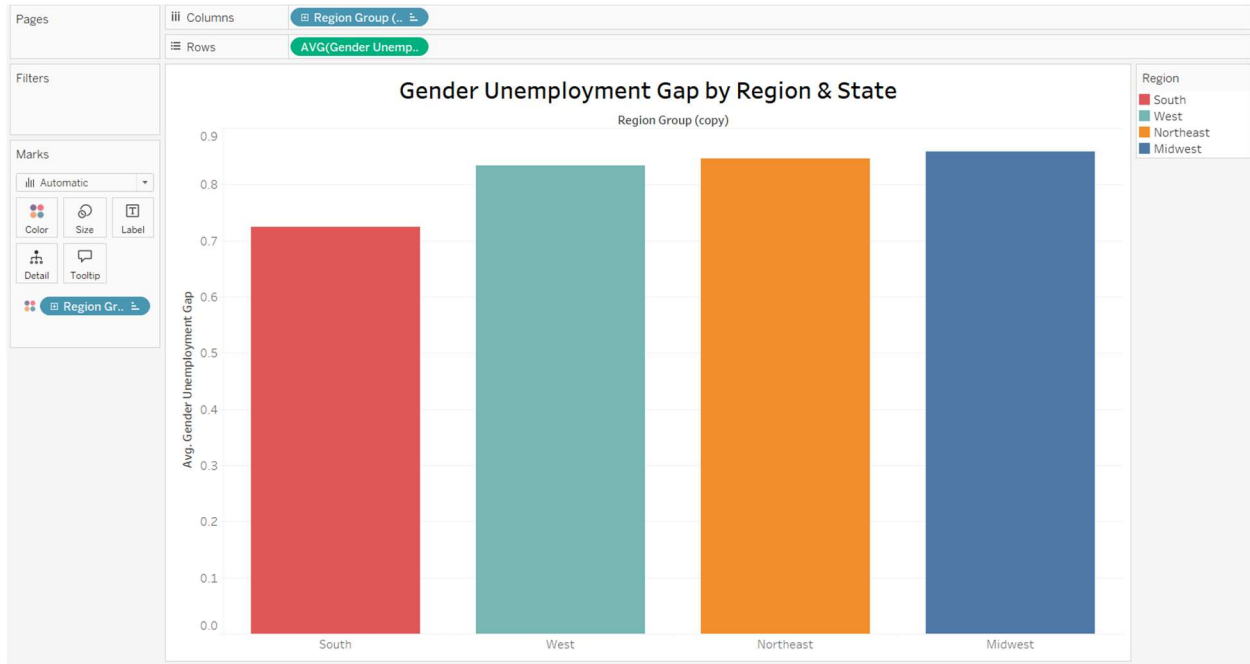
○ National Trends in Crime (2010–2019)



The visualization shows that violent crime rates remained relatively stable nationwide between 2010 and 2019, with both the annual and moving average values indicating only minor fluctuations across the decade. Property crime rates, however, exhibited periods of change, including a sharp decline between 2012 and 2014. This decline is evident in both the annual and moving averages, with the annual figures showing a steeper drop. The data also indicates relative increases in property crime in the early part of the decade and again between 2017 and 2019. These trends address the question by illustrating that while violent crime rates remained steady, property crime rates experienced more variability during the same period.

2. How does the gender unemployment gap differ by state and region?

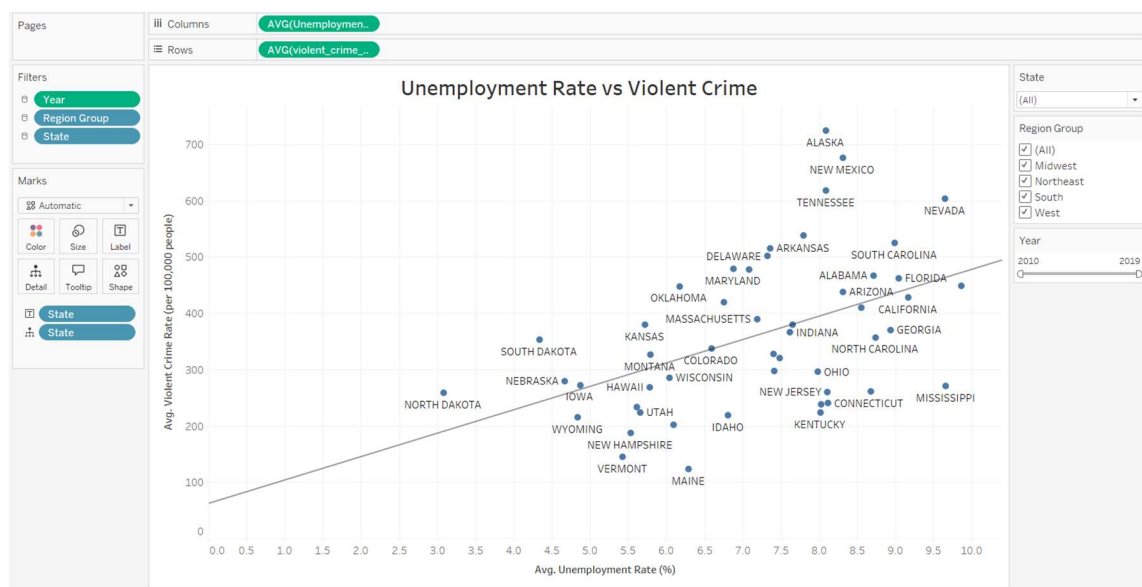
Gender Unemployment Gap by Region and State



The gender unemployment gap across the U.S. between 2010 and 2019 shows clear differences by both region and state. The South stands out for having the lowest average gap, with most states in the region showing similarly low values, suggesting more consistent gender parity in unemployment outcomes. The Midwest and Northeast, by contrast, have higher and more uniform gaps across states, pointing to broader regional patterns of inequality. The West is more mixed: while it includes both the state with the smallest gap (Wyoming) and the largest (Alaska), most other Western states fall somewhere in between. This highlights an important point, the fact that a single regional average can hide a lot of variation underneath. Ultimately, the chart reveals that gender unemployment gaps tend to be smaller and more consistent in the South, while the Midwest and Northeast experience higher and more uniform disparities, and the West displays greater internal variation. This demonstrates that gender unemployment inequality in the U.S. differs not only in degree but also in distribution across both regions and individual states.

3. Is there a relationship between unemployment rates and violent crime rates across states?

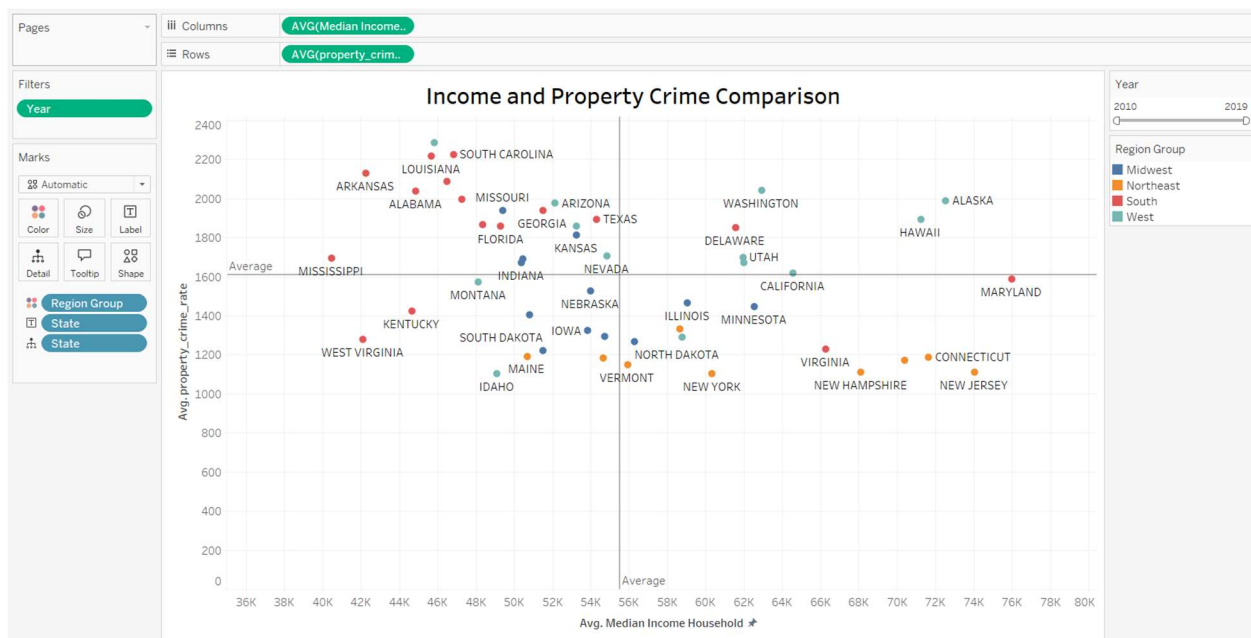
○ Unemployment Rate and Violent Crime Correlation



The visualization suggests a weak positive relationship between unemployment rates and violent crime rates across U.S. states from 2010 to 2019. States with higher average unemployment tend to report somewhat higher levels of violent crime, though the association is not especially strong or consistent. Several states with similar unemployment rates display very different crime levels, and vice versa, indicating that while economic conditions may be one contributing factor, they do not appear to be the sole or dominant driver of violent crime. The trend line implies a general upward pattern, but the wide spread of data points shows considerable variation. Overall, the chart indicates that there is a modest correlation between unemployment and violent crime across states, but it is not strong enough to suggest a clear or uniform relationship.

4. How do median household income levels relate to property crime rates across U.S. states?

Income and Property Crime Comparison

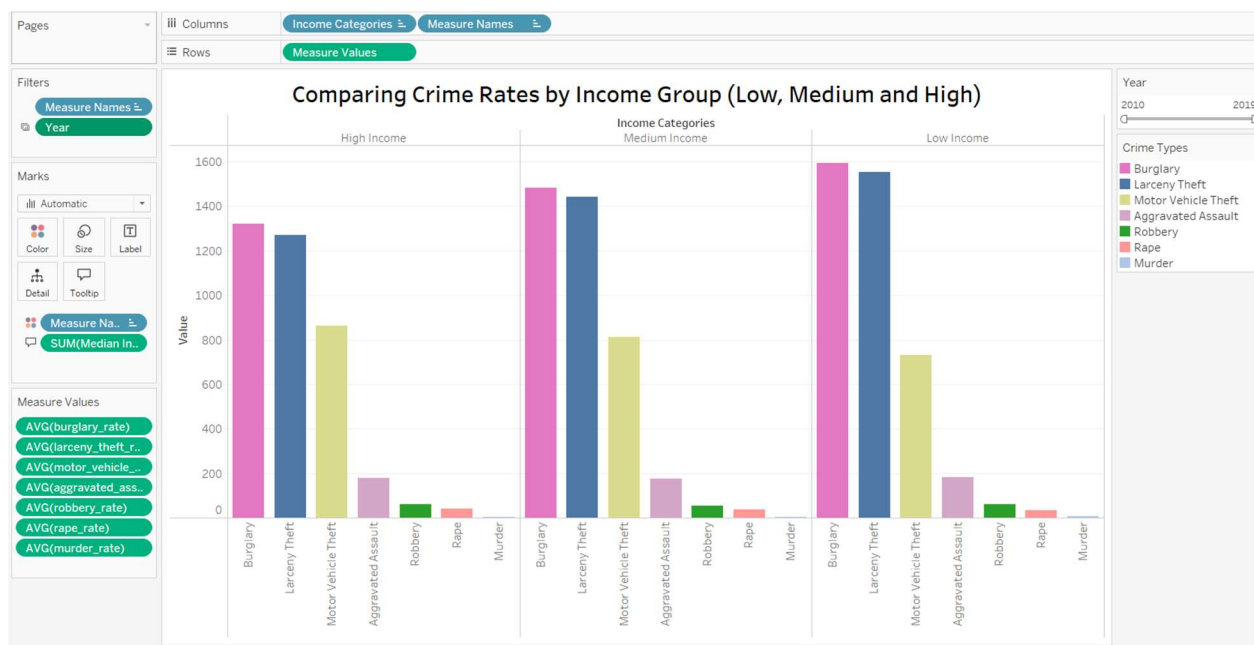


While the data points are somewhat dispersed, a general pattern emerges: states with lower median household incomes tend to experience higher property crime rates, whereas higher-income states tend to report lower crime. However, this relationship is modest and not

universally consistent. Some high-income states, such as Alaska and Hawaii, still exhibit elevated property crime levels, indicating that income alone does not fully explain the variation. When viewed by region, additional patterns become visible: many Southern states cluster in the lower-income, higher-crime quadrant, while Northeastern states often appear in the higher-income, lower-crime area. These patterns suggest that while income level is a contributing factor, regional context and other underlying factors also influence property crime rates across U.S. states.

5. How do overall crime rates vary between states with different income levels (low, medium, and high)?

○ Comparing Crime Rates by Income Group

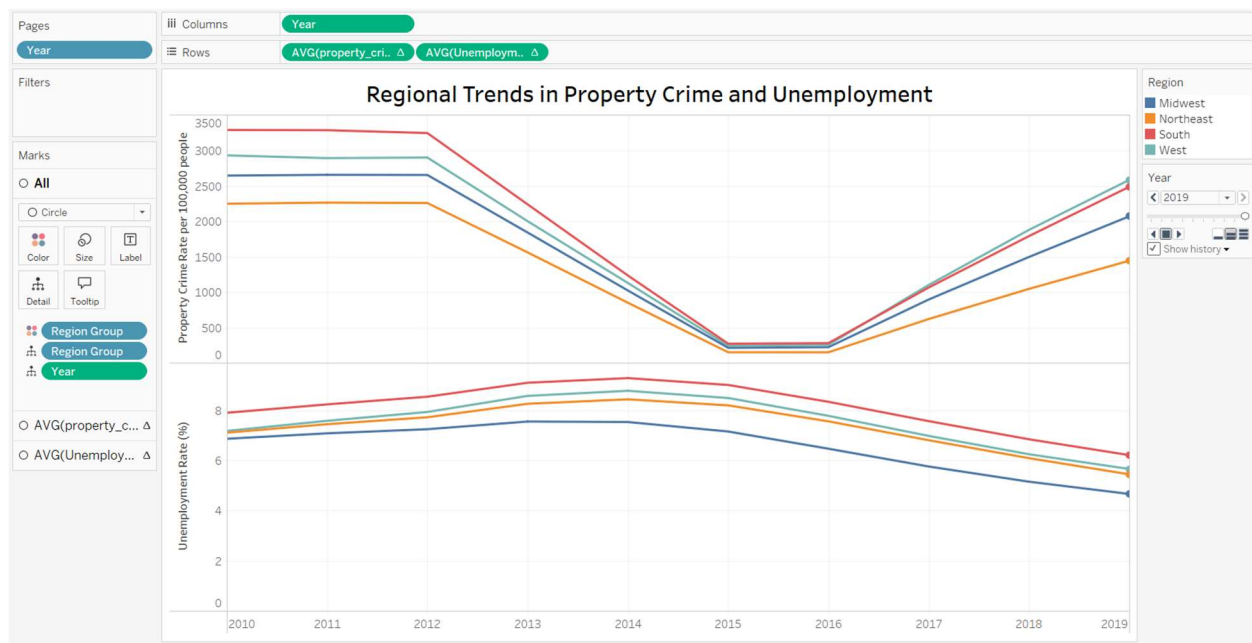


The data reveals that overall crime patterns vary by income level, but not uniformly across all crime types. Property crimes such as burglary and larceny-theft are most prevalent in low-income states and decrease as income levels rise. However, motor vehicle theft shows a different

pattern, with the highest rates in high-income states, followed by medium- and low-income groups. In terms of violent crimes—including aggravated assault, robbery, rape, and murder—rates are relatively consistent across income levels, with only slight differences between groups. These findings suggest that while income level is strongly associated with certain types of property crime, its relationship with violent crime is less clear and more complex.

6. How do property crime and unemployment trends vary across U.S. regions over time (2010–2019)?

○ Regional Trends in Property Crime and Unemployment

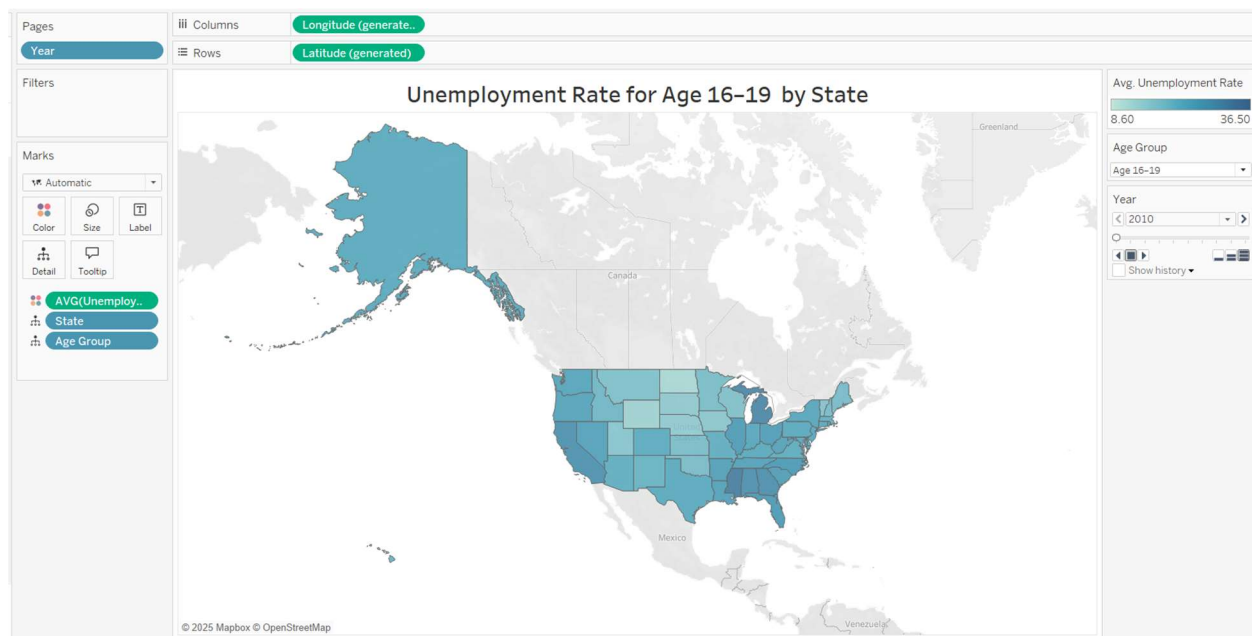


At a glance, the data reveals noticeable shifts in both property crime and unemployment across U.S. regions from 2010 to 2019. Unemployment rates across all regions initially rose, peaking around 2013–2014, before steadily declining through 2019. Property crime rates remained relatively stable between 2010 and 2012, then declined significantly from 2012 to 2015, followed by a period of minimal change. Throughout the decade, the South consistently recorded

the highest property crime rates, with the West also elevated, while the Northeast remained the lowest. Unemployment trends were more tightly grouped across regions, though the Midwest generally maintained the lowest rates. This shows that while national trends point to an overall decline, regional variation remained consistent throughout the decade, with the South and West experiencing persistently higher crime and unemployment compared to the Northeast and Midwest.

7. How do unemployment rates vary across U.S. states and over time for different age groups?

○ Unemployment Rate by Age Group by State and Year



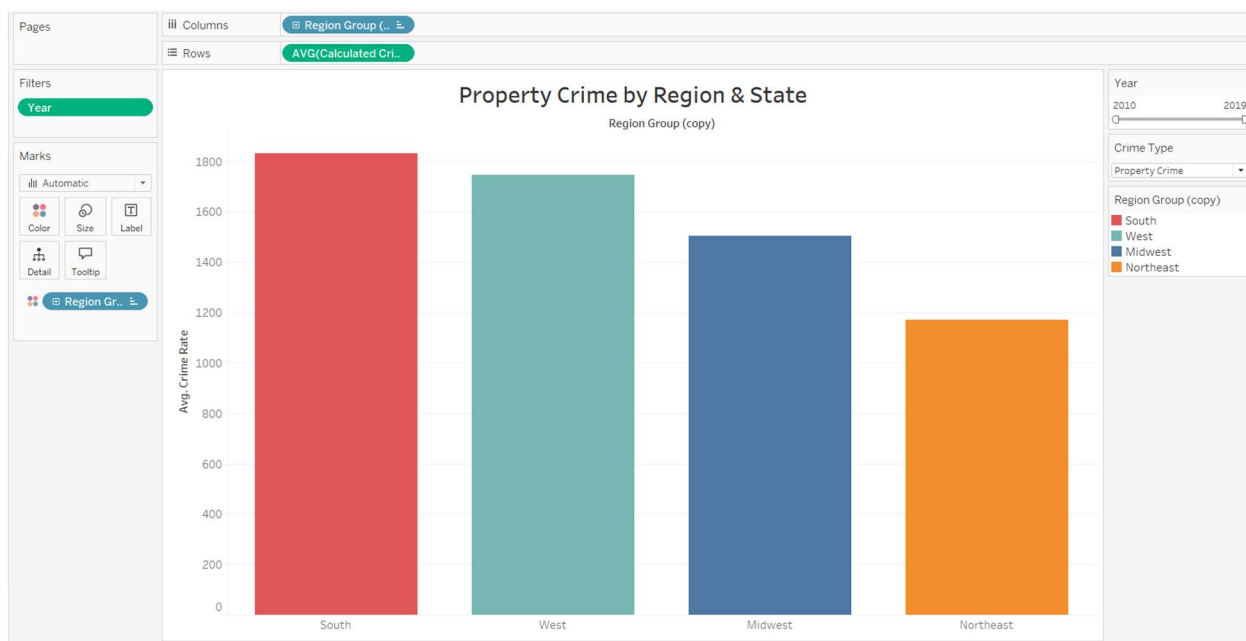
Unemployment rates vary notably across U.S. states and age groups, with younger populations generally experiencing higher unemployment than older adults. The 16–19 age group consistently shows elevated unemployment levels across most states and over time, reflecting their more precarious position in the labor market. In contrast, age groups such as 25–44 tend to

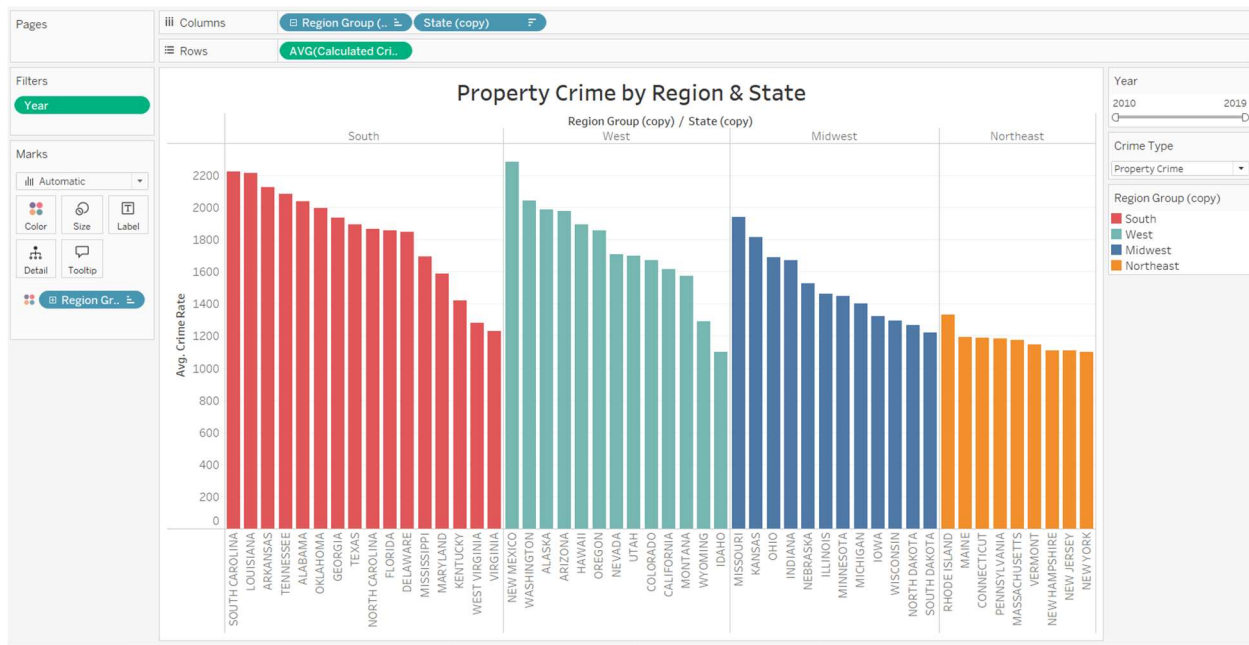
exhibit lower and more stable unemployment rates. However, the degree of difference between age groups is not uniform, some states and time periods show a sharper gap than others. Overall, the visualization highlights age as a key factor influencing unemployment, with younger individuals facing greater challenges in securing employment across regions and years.

8. How do average violent and property crime rates differ across U.S. regions and states?

○ Crime Rate by Region and State

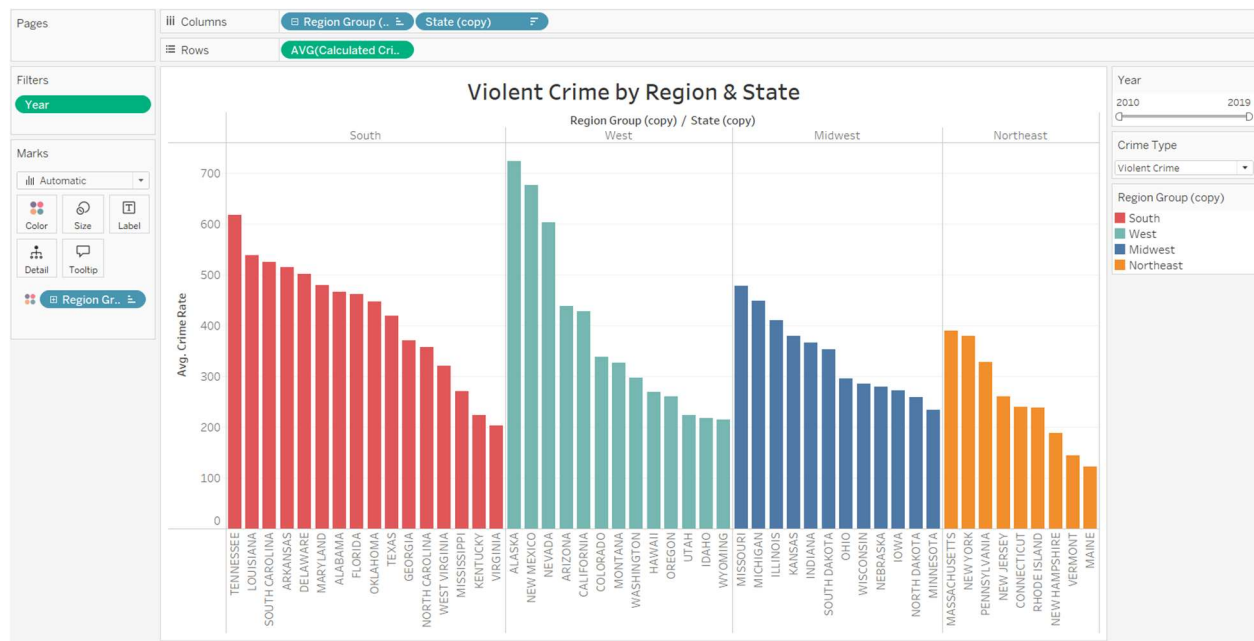
Crime Type: Property Crime





Crime Type: Violent Crime

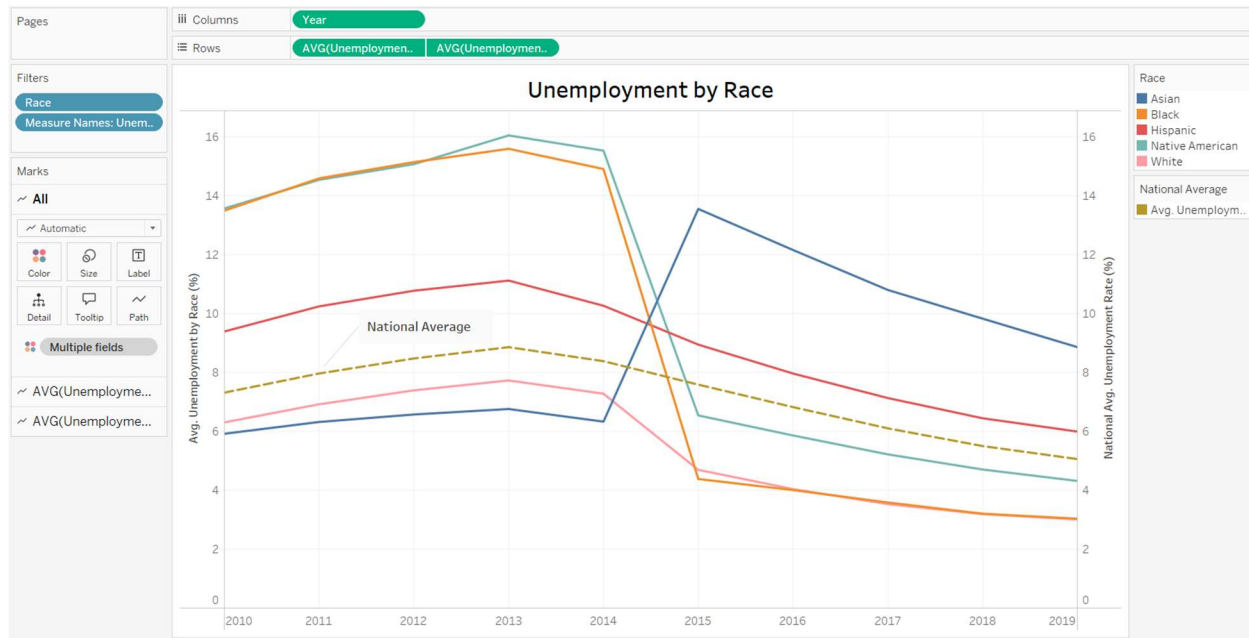




Both violent and property crime rates vary substantially across U.S. regions and states, with consistent patterns emerging across crime types. The South reports the highest average crime rates overall, followed by the West, while the Midwest and especially the Northeast show significantly lower levels. This regional pattern holds for both violent and property crimes, although the differences are more pronounced for property crime. At the state level, there is considerable variation within regions—for example, some Western states exhibit particularly high violent crime rates despite the region's moderate average. These findings suggest that while broad regional trends exist, state-level differences can be substantial, particularly in states with outlier crime rates. Overall, the South and West face higher crime burdens on average, while the Northeast remains the lowest across both categories.

9. How do unemployment trends differ by race over time, compared to the national average?

○ Unemployment by Race



This time-series line chart tracks unemployment rates across five racial groups - Asian, Black, Hispanic, Native American, and White - compared to the national average. In the early part of the decade, Black and Native American populations had the highest unemployment rates, while Asian and White groups remained well below the national average. Around 2014, these trends shifted: unemployment declined notably for Black and Native American groups, while Asian unemployment rose above the national average and remained elevated in the years that followed. Hispanic unemployment stayed consistently higher than the national trend across the entire period. Overall, the chart highlights that unemployment patterns differ significantly by race and shift over time, some groups remain consistently disadvantaged, while others move above or below the national average at different points.

10. What types of violent crimes are most common across U.S. regions?

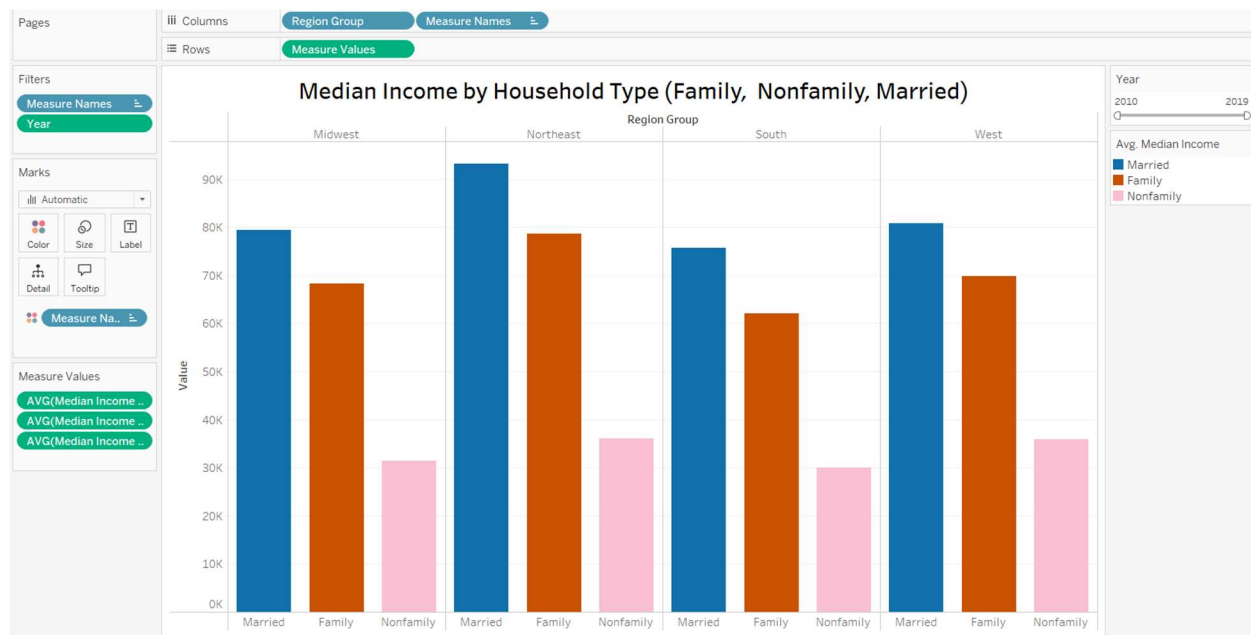
○ Violent Crimes by Region



This bar chart compares the average rates of four major violent crimes - Aggravated Assault, Robbery, Rape, and Murder, across the West, South, Midwest, and Northeast regions. Aggravated Assault stands out as the most prevalent violent crime in every region, particularly in the South and West. Robbery and Rape follow in frequency, with Robbery being slightly higher in most regions. Murder rates are the lowest across all regions, showing relatively little variation. Overall, the visualization reveals that while violent crime exists nationwide, the South and West experience notably higher levels, and Aggravated Assault is consistently the dominant form across the board.

11. How does household type affect earnings across regions?

○ Median Income by Household Type (Family, Nonfamily, Married)



This visualization explores how household structure relates to economic outcomes across U.S. regions. Married households consistently show the highest income levels, followed by family households, with nonfamily households trailing significantly behind. This gap is visible across all regions and reveals a structural advantage associated with marriage or multi-person family units. The consistent underperformance of nonfamily households suggests that individuals living alone or in nontraditional arrangements may be at a structural disadvantage when it comes to economic security. Overall, the chart shows that household type has a clear and consistent effect on income across regions, with married households maintaining a distinct financial advantage.

Dashboards

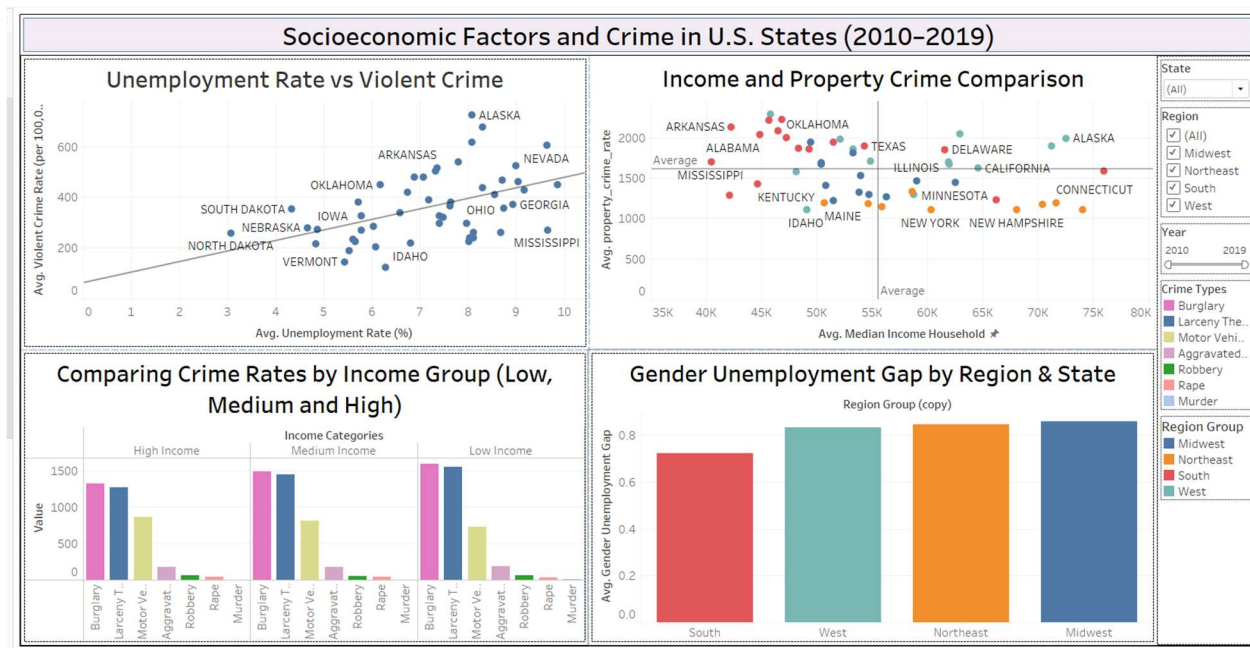
To explore key socioeconomic patterns and their possible connections to crime, three dashboards were developed. These visualizations examine a range of factors, including unemployment,

income levels, race, household type, age, and gender, across U.S. states and regions from 2010 to 2019. While some dashboards highlight direct relationships with crime, others focus on broader demographic and economic disparities to provide deeper context for understanding regional differences.

Dashboard 1: Socioeconomic Factors and Crime in U.S. States (2010–2019)

This dashboard explores how crime correlates with key socioeconomic variables such as unemployment, income level, and gender-based employment disparities. The visualizations collectively address the research question:

How do socioeconomic factors like unemployment, income level, and gender disparities relate to crime rates across states?



It enables users to:

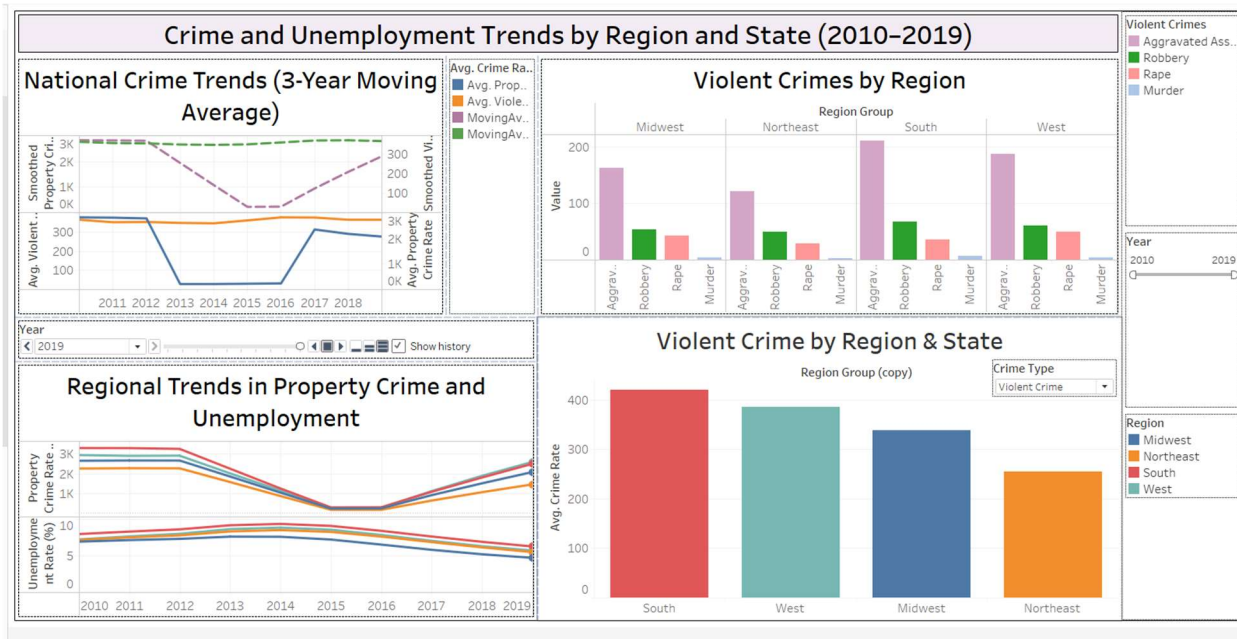
- Assess the correlation between unemployment and violent crime.

- Explore the relationship between median income and property crime rates.
- Compare how crime types vary across income categories (low, medium, and high).
- Understand gender disparities in unemployment by region and state.

Dashboard 2: Trends in Crime and Socioeconomic Indicators (2010–2019)

This dashboard examines regional variations in violent crime and the evolution of property crime and unemployment rates over time. It addresses the research question:

How do crime and unemployment trends differ across U.S. regions and states from 2010 to 2019?



It allows users to:

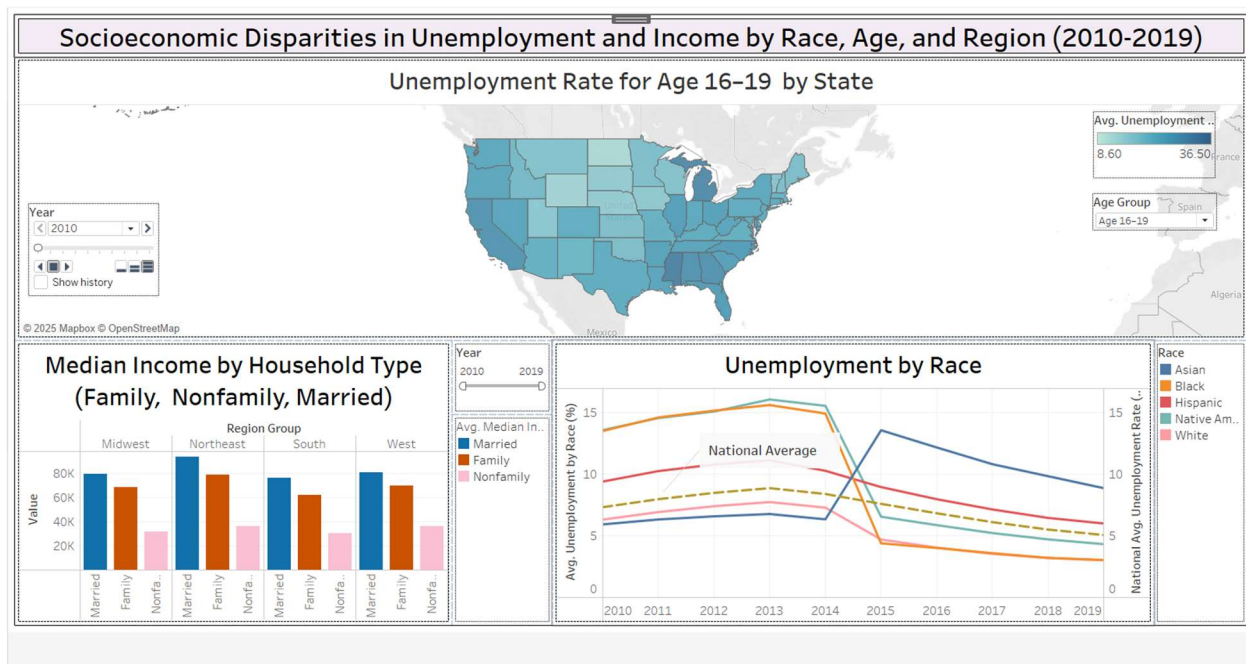
- Analyze national trends in violent and property crime using a 3-year moving average.
- Explore which violent crimes are most prevalent in different regions.
- Compare average property and violent crime rates by region and state.

- Explore relationship between unemployment and property crime by region over the decade.

Dashboard 3: Unemployment, Crime, and Income Disparities Across U.S. Regions (2010-2019)

This dashboard focuses on economic inequalities tied to race, age, and household structure. The visualizations collectively address the research question:

How do unemployment and income levels vary by race, age group, and household type across U.S. regions?



It enables users to:

- Compare unemployment trends by race alongside the national average.
- Understand how household type influences income levels across regions.
- Explore geographic variation in youth (age 16-19) unemployment across states.

Together, the three dashboards reveal clear patterns related to income, unemployment, gender, race, household type, and geography, while also highlighting important differences across states, regions, and over time. These insights collectively help address the overarching research question:

How do socioeconomic conditions and demographic factors influence crime patterns across U.S. states and regions from 2010 to 2019?

4. Conclusion

This project explored how socioeconomic and demographic factors - such as unemployment, income, gender, and age - relate to crime across U.S. states and regions from 2010 to 2019.

Through a series of interactive visualizations, several key patterns emerged: property crime rates tend to be higher in low-income states, unemployment among youth and women shows notable regional variation, and regional differences persist in both crime and employment trends. While some relationships, like income and property crime, appear stronger, others, such as the link between violent crime and unemployment, are more complex and inconsistent.

However, the analysis also highlights the challenges of drawing definitive conclusions from observational data. Both crime and unemployment are influenced by a wide range of structural, economic, and policy-related factors not fully captured here. Additionally, certain patterns vary significantly across states and years, underscoring the need for localized or more time-sensitive research.

Looking ahead, future research could explore questions such as:

1. How do education levels or access to services relate to crime outcomes across states?

2. Are there measurable lag effects between changes in unemployment and shifts in crime rates?
3. How do structural differences across states (e.g., urbanization, policy) influence these patterns?

By expanding the scope and granularity of future analysis could help researchers and policymakers better understand the interplay between social inequality and crime, and ultimately inform more targeted and effective interventions.